THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Yes	Digit sum 18 is divisible by 9
2	**(b)** Yes	Difference of sums of alternate digits
3	**(b)** 2	Digit sum must be a multiple of 3
4	**(d)** 0	Digit sum must be a multiple of 9
5	**(c)** 75	Smallest common multiple
6	**(b)** 2475	$(100-1)\times 25$ using the distributive property
7	**(c)** 961	Use $(a\pm b)^2$
8	**(c)** 10609	Use $(a\pm b)^2$
9	**(b)** 2000	$(a+b)(a-b)$
10	**(d)** 81	$a^2+2ab+b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Arrange digits in descending and ascending order, subtract; repeat to reach 495 (2).	2
12	Every other even number has 2 as a factor besides 1 and itself (2).	2
13	$x^2+8x+15$ (2).	2
14	x^2-16 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Even and digit sum 21 is divisible by 3 (2); so divisible by 6 (1).	2+1
16	$x^2+7x+12$ (2); at $x=1$: $(4)(5)=20=1+7+12$ (1).	2+1
17	5432−2345=3087 (1); 8730−0378=8352 (1); 8532−2358=6174 (1).	1+1+1
	OR: Alternate sums: $3+5$ and $\square+2$ must differ by 0 or 11, so $\square=6$ (3).	3
18	$x^2+12x+35$ (2); at $x=1$: $(6)(8)=48=1+12+35$ (1).	2+1
	OR: $x^2+11x+18$ (2); at $x=1$: $(3)(10)=30=1+11+18$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) One is even and one is a multiple of 3 (3). (ii) $504=6\times 84$ (2).	3+2
	OR: (i) 741−147=594 (2). (ii) 954−459=495 (3).	2+3
20	(i) Expansion (3). (ii) $4\times 15\times 9=540$ (2).	3+2
	OR: (i) Statements with numbers (3). (ii) Commutative and associative: $18\times(25\times 4)=1800$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	Rule for 4
2	**(a)** 3475	Ends in 0 or 5
3	**(d)** 495	Kaprekar for 3 digits
4	**(d)** 12	Greatest common factor
5	**(c)** 111111111	A curious pattern
6	**(b)** $a^2 - 2ab + b^2$	Standard identity
7	**(d)** associative property	Grouping does not matter
8	**(c)** 7	$5 + 2$
9	**(d)** 700	$(25 \times 4) \times 7$
10	**(d)** non-negative	A square is never negative

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $10 + x$ must be 18, so $x = 8$ (2).	2
12	$A = 5$ (2).	2
13	$15(27 + 73) = 1500$ (2).	2
14	$2500 + 200 + 4 = 2704$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM $= 36$ (2); at 8:36 a.m. (1).	2+1
16	$x^2 + 14x + 48$ (2); at $x = 1$: $(7)(9) = 63 = 1 + 14 + 48$ (1).	2+1
17	$777 = 3 \times 259 = 3 \times 7 \times 37$ (3).	3
	OR: Number $10a + b$, reverse $10b + a$, difference $9(a - b)$ (3).	3
18	$x^2 + 15x + 44$ (2); at $x = 1$: $(5)(12) = 60 = 1 + 15 + 44$ (1).	2+1
	OR: The middle term $10x$ is missing (1.5); correct is $x^2 + 10x + 25$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1001, 1008, …, 1099 gives 15 (2). (ii) 1001, 1012, …, 1100 gives 10 (2). (iii) 1001 and 1078 (1).	2+2+1
	OR: (i) 854−458=396; 396+693=1089 (3). (ii) 732−237=495; 495+594=1089, so always 1089 (2).	3+2
20	(i) The cross terms $ab + ab$ are lost (2). (ii) $25 \neq 17$ (2). (iii) $a^2 + 2ab + b^2$ (1).	2+2+1
	OR: (i) $x^4 - 4$ (3). (ii) $77 = 11 \times 7$ (2).	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Yes	Digit sum 18 is divisible by 9
2	**(b)** Yes	Difference of sums of alternate digits
3	**(a)** 3	Digit sum must be a multiple of 9
4	**(c)** 1	Digit sum must be a multiple of 3
5	**(b)** 42	Smallest common multiple
6	**(b)** 1751	$(100+3)\times 17$ using the distributive property
7	**(d)** 2401	Use $(a\pm b)^2$
8	**(a)** 9409	Use $(a\pm b)^2$
9	**(a)** 4400	$(a+b)(a-b)$
10	**(d)** 169	$a^2+2ab+b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	100 (1); the sum of first n odd numbers is n^2 (1).	1+1
12	$1001 = 7\times 11\times 13$, so yes to all (2).	2
13	$400-40+1=361$ (2).	2
14	$a^2-2ab+b^2$ (1); $9=25-20+4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Even, so divisible by 2 (1); digit sum 27, so divisible by 3 and 9 (1); last two digits 18 are not divisible by 4 (1).	1+1+1
16	$(a^2+2ab+b^2)-(a^2-2ab+b^2)=4ab$ (3).	3
17	LCM 360 (1.5); HCF 12 (1.5).	1½+1½
	OR: LCM $=120$ (2); 123 (1).	2+1
18	$16+4+0.25=20.25$ m² (3).	3
	OR: $6x^2-15x+4x-10=6x^2-11x-10$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Last digit △ is 0 or 5 (2). (ii) Digit sum $8+\square+\triangle$ is a multiple of 9, giving 5130 and 5535 (3).	2+3
	OR: (i) Sum of its proper divisors equals the number (2). (ii) $1+2+3=6$; $1+2+4+7+14=28$ (3).	2+3
20	(i) $100\times 72/72=100$ (2.5). (ii) $100\times 16/16=100$ (2.5).	2½+2½
	OR: (i) $11^2-9^2=40$ (2). (ii) 50^2-49^2 and 18^2-15^2 (3).	2+3

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	Rule for 8
2	**(c)** even	odd + odd = even
3	**(b)** the same forwards and backwards	e.g. 12321
4	**(b)** 36	Coprime factors
5	**(d)** 28	28 = 1+2+4+7+14
6	**(d)** $a^2 - b^2$	Difference of squares
7	**(a)** 1	$a \times 1 = a$
8	**(c)** 100	10^2
9	**(b)** 6	$-3 \times (-2)$
10	**(a)** 4	$xp + xq + yp + yq$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum 11, then 2; remainder $= 2$ (2).	2
12	Yes; 3 and 4 are co-prime and $3 \times 4 = 12$ (2).	2
13	Adding 0 to any number leaves it unchanged (2).	2
14	$m^2 - n^2$ (1); difference of two squares (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum of digits at odd places (from the right) $= 1$, at even places $= 2$ (2); difference 1 is not a multiple of 11, so no (1).	2+1
16	$9m^2 - 12mn + 4n^2$ (2); $36 - 24 + 4 = 16$ (1).	2+1
17	Digit sum 22 leaves remainder 4 (1.5); $4729 = 9 \times 525 + 4$ (1.5).	1½+1½
	OR: 14, 28, 42, 56, 70 (2); 28 and 56 (1).	2+1
18	$x^2 + 3x - 10$ (2); 44 (1).	2+1
	OR: Take $a = 5, b = 2$: $9 \neq 21$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Primes differing by 2: (3,5), (5,7), (11,13), (17,19), (29,31), (41,43) (3). (ii) $91 = 7 \times 13$, not prime (2).	3+2
	OR: (i) 8:00, 8:15, 8:30, 8:45 (1). (ii) LCM of 15 and 20 is 60 minutes (2). (iii) At 8:00 and 9:00 (2).	1+2+2
20	(i) $36 = 1 + 4 + 9 + 4 + 12 + 6$ (3). (ii) 169 (2).	3+2
	OR: (i) $23 \times 17 = 391$ m² (2). (ii) $400 - 391 = 9$ m² less (1). (iii) $(a + b)(a - b) = a^2 - b^2$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.